Blind Speech-Enhancement Floors under Harmonic Rotor Noise

A five-architecture, two-pass baseline study on drone and harmonic noise

Dmitrii Mukhutdinov[a]

[a] Harmonic Noise Suppression Project

Abstract

We establish the **blind** (no rotor-speed side information) speech-enhancement floor on our ultra-low-SNR (−30…0 dB) harmonic-noise data with five modern architectures, trained in two passes — on drone noise only, and on a category-uniform mix of nine harmonic-noise families — and scored on a fixed, held-out validation set against noisy-input and Wiener anchors. The floor is strongly **architecture-bound**: MP-SENet (parallel magnitude+phase) is the strongest baseline, and both ported models (MP-SENet, TF-GridNet) beat the in-house Edge-BS-RoFormer — even while compute-limited — while the classic complex-UNet (DCUNet) only removes noise energy without restoring intelligibility. Whether training on **diverse** harmonic noise helps on drone noise is **capacity-dependent** — it aids only the capacity-limited DCUNet and dilutes all three stronger ports (MP-SENet, TF-GridNet, Edge-BS-RoFormer). These floors gate later rotor-informed claims.

Keywords: speech enhancement; drone ego-noise; ultra-low SNR; SI-SDR; blind baselines.

Contents

Introduction 3
Methods 3
  Data and mixing 3
  Architectures 3
  Loss and training 3
  Evaluation 3
Results 4
Discussion 5
  Status and caveats 5
Bibliography 6

Introduction

Speech enhancement under the harmonic ego-noise of rotating sources (drone motors and propellers) is an extreme low-SNR problem: at onboard microphones the target speech sits −30 to 0 dB below dense, structured rotor noise. Before any rotor-speed-informed method can claim a benefit, we need an honest **blind** floor — how well modern architectures do with no side information — measured on our own data with strong anchors. This report establishes that floor.

We ask two questions. **(i)** Which modern architectures set the blind floor, and by how much do they beat trivial anchors (noisy input, Wiener)? **(ii)** Does training on a **diverse** set of harmonic noises help on drone noise (transferable harmonic structure) or hurt (capacity dilution)? We answer both with a five-architecture set trained in two passes and evaluated per input SNR.

Methods

Data and mixing

Speech is LibriSpeech `train-clean-100` (25 speakers held out for validation). Noise is drawn online each step and mixed at SNR $\sim \mathcal{U}(-30, 0)$ dB with `random_gain`/`random_polarity` augmentation, 16 kHz mono, 1 s chunks. Two noise regimes define the passes:

- **Pass A (drone only):** DREGON + Michael's real drone recordings plus open drone-noise datasets, roughly uniform over sub-datasets.
- **Pass B (all harmonic):** category-uniform over nine families — drone, MIMII (industrial), MIMII-DG, aircraft (AeroSonicDB), motors (HUSTmotor, KAIST), horns (HornBase) — so the 258 GiB MIMII cannot dominate.

The noise pool streams lazily from object storage at shard granularity, capped at 24 shards per dataset (a bounded, diverse subset — an uncapped random-shard-per- sample draw would stream the whole 258 GiB MIMII over a run). Both passes are scored on the **same** fixed, published validation set: held-out noise recordings (recording-level split) × held-out speakers, SNR grid $\{-30, -25, -20, -15, -10, -5, 0\}$ dB, 50 mixtures/point, deterministic. Every table carries **noisy-input** and **Wiener** (`scipy.signal.wiener`) anchors.

Architectures

Five blind (no-RPS) waveform-in/waveform-out models: **Edge-BS-RoFormer** (band-split rotary transformer, our Paper-1 model), **TF-GridNet** (dense full+sub-band dual-path, mid-size $\sim$ 8.4 M, ported), **MP-SENet** (parallel magnitude+phase, $\sim$ 1.7 M, ported), **DCUNet** (complex U-Net, the 2023 benchmark winner [1] — a continuity anchor), and **SGMSE+** (score-based diffusion, ported, trained from scratch; deferred; see the caveats below).

Loss and training

An initial masked-MSE pass produced floors **at** the noisy level (at ultra-low SNR, MSE rewards attenuation-toward-silence, improving SDR but not intelligibility). We therefore train with a metric-aligned composite: negative SI-SDR + a multi-resolution STFT term. Training length matches the 2023 benchmark [1]: early-stop patience $N_E = 30$, LR-plateau patience $N_\alpha = 15$, $\sim$ 1300 steps/epoch, cap 150 epochs, AdamW at 10^{-3}, `amp` off (complex STFT). Jobs run on A100 GPUs.

Evaluation

The single evaluation path scores each checkpoint per SNR with SI-SDR, SDR, PESQ (wideband) and extended STOI. For a **fair** cross-model comparison all models are scored on the same balanced 25-mixtures/SNR subset (SI-SDR is high-variance at 0 dB, so mixing 25- and 50-clip means across models is unfair; the 25-subset ranking is stable, absolute values at 0 dB carry $\pm$ a few dB).

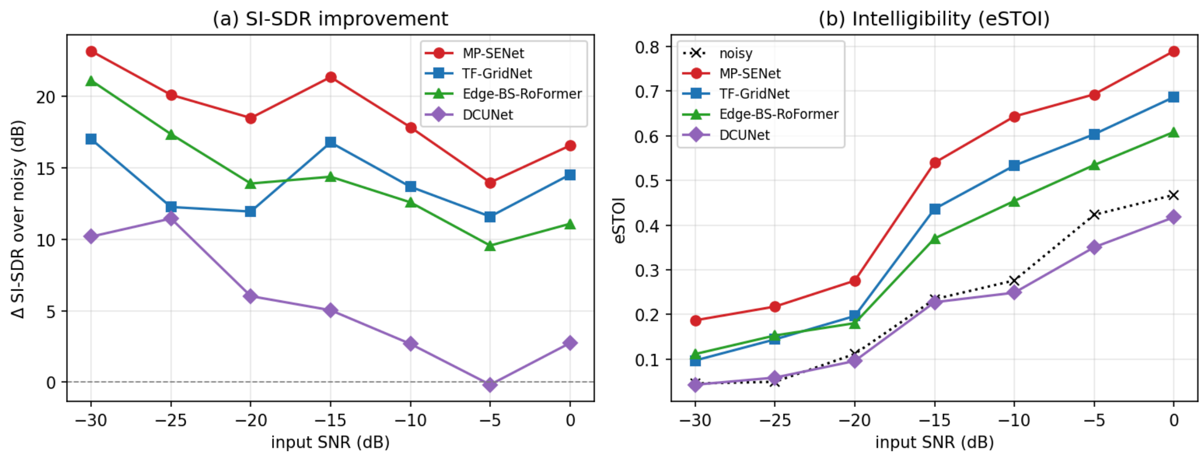

Figure 1. Blind SE floor on `SE-valid-drone` (Pass A). (a) SI-SDR improvement over the noisy input vs input SNR. (b) Absolute eSTOI (intelligibility); the dotted line is the noisy input. MP-SENet and TF-GridNet — both ported — lead the in-house Edge-BS-RoFormer; DCUNet tracks the noisy eSTOI.

Table 1. SI-SDR improvement over the noisy input (dB), per input SNR, uniform 25-mixtures/point. Bold = best. Every model clears the Wiener anchor at ≥ -10 dB.

SNR	Wiener	**MP-SENet**	TF-GridNet	Edge-BS-RoF	DCUNet-A	DCUNet-B
−30	−0.1	**+23.2**	+17.0	+21.1	+10.2	+11.9
−20	+0.1	**+18.5**	+11.9	+13.9	+6.0	+6.0
−15	+0.1	**+21.4**	+16.8	+14.4	+5.0	+7.6
−10	+0.1	**+17.9**	+13.7	+12.6	+2.7	+3.9
−5	+0.1	**+14.0**	+11.6	+9.6	−0.2	+0.7
0	+0.1	**+16.6**	+14.5	+11.1	+2.7	+2.2

Table 2. Absolute eSTOI (higher = more intelligible), noisy input vs models.

SNR	noisy	**MP-SENet**	TF-GridNet	Edge-BS-RoF	DCUNet
−15	0.234	**0.540**	0.436	0.370	0.228
−10	0.276	**0.643**	0.533	0.454	0.249
−5	0.423	**0.693**	0.603	0.534	0.350
0	0.468	**0.789**	0.687	0.609	0.418

Results

Figure 1 and Table 1 summarise Pass A on the drone validation set. The spread across architectures is large and the ranking is unambiguous: **MP-SENet > TF-GridNet ≳ Edge-BS-RoFormer ≫ DCUNet**.

Intelligibility (eSTOI). The models separate most clearly on eSTOI (Table 2). MP-SENet nearly doubles noisy intelligibility at −15 dB (0.234 ⟶ 0.540) and reaches 0.789 at 0 dB; TF-GridNet and Edge-BS-RoFormer follow; DCUNet stays at the noisy level — it removes noise energy (SI-SDR/PESQ rise) without restoring speech.

Diversity verdict — capacity-dependent (Table 3). We compare each model's drone-only pool (Pass A) against the all-harmonic pool (Pass B), scored on the drone valid (both passes select the best drone-valid checkpoint, and both are the same compute-limited regime, so the delta is a fair within-

Table 3. Diversity delta: Δ SI-SDR of Pass B (all-harmonic) minus Pass A (drone-only) on the drone valid (dB). Positive (bold) = diversity helps. It helps only the capacity-limited DCUNet; the three stronger ports are hurt or unmoved.

model (B−A)	−30	−25	−20	−15	−10	−5	0
DCUNet (weak)	**+1.7**	−0.0	−0.0	**+2.5**	**+1.2**	**+0.9**	−0.6
MP-SENet	−3.6	−4.6	−4.3	−2.6	−1.4	−0.1	+0.0
TF-GridNet	−1.1	−0.8	−1.3	−2.2	−3.6	−2.3	−1.3
Edge-BS-RoF	−3.6	−4.5	−6.3	−4.4	−2.9	−3.0	−1.3

architecture comparison). The verdict splits cleanly by capacity: for the weak **DCUNet**, Pass B $\geq$ Pass A at most SNRs (Δ SI-SDR at −15 dB: +2.5) — extra harmonic data helps a model that under-fits drone noise alone. But **all three stronger ports are hurt or unmoved**: Edge-BS-RoFormer and TF-GridNet lose SI-SDR at every SNR and eSTOI across the board (Edge-BS-RoF eSTOI −0.036 at −15 dB, TF-GridNet −0.031), while MP-SENet loses SI-SDR at low SNR and is flat on eSTOI. Under an equal training budget, diverse noise dilutes drone-specific capacity; the transferable structure of rotating-source harmonics helps only when the model is capacity-limited on the target. A capable model does better **focusing** on the target noise.

Loss and length matter. On DCUNet, masked-MSE gave a floor essentially at the noisy level; the SI-SDR+MRSTFT composite with the paper-matched schedule lifted the −15 dB SI-SDR gain from +1.6 dB (an early-stopping-truncated 12-minute run) to +5.0 dB — a reminder that a “baseline” is a claim about training as much as architecture.

Discussion

The blind floor on our data is set by the **newer** speech-enhancement architectures: MP-SENet’s explicit magnitude-and-phase decoders are the strongest, and both ports beat the in-house Edge-BS-RoFormer. This is a useful recalibration — the strongest blind baseline is not the band-split transformer we had been treating as SOTA. The intelligibility gap between models that **restore** speech (MP-SENet, TF-GridNet, Edge-BS-RoFormer) and one that merely **denoises** (DCUNet) is the headline: at these SNRs, SI-SDR and PESQ can rise while eSTOI does not, so intelligibility metrics are the ones that separate real enhancement from attenuation.

That MP-SENet and TF-GridNet win **while compute-limited** (the caveats below) means their true ceiling is higher still. The diversity result cautions against a one-size-fits-all data recipe: broadening training to many harmonic-noise families helps only the capacity-limited DCUNet and **hurts or fails to help** all three stronger ports on the target noise. The split is clean across every model we ran, so data breadth should scale with — not substitute for — architecture and training budget.

Status and caveats

The MP-SENet and TF-GridNet numbers are **lower bounds**: `amp`-off fp32 makes them $\sim 10\times$ slower per step than DCUNet (MP-SENet 1.5 it/s; TF-GridNet slower at batch 4), so both hit the wall-clock wall before convergence; their best checkpoints were recovered from object storage. fp16 training (STFT kept in fp32) would let them converge and would likely widen their lead. The Pass B deltas for the compute-limited MP-SENet and TF-GridNet are read from their best-drone-valid checkpoints while their runs continue, matched against the same-regime Pass A checkpoints; the direction of the diversity effect is stable, absolute magnitudes may tighten as they converge. SGMSE+ (65 M, score-matching) is deferred — it needs a bespoke training loop and a multi-day budget. The per-category **harmonic-noise** transfer table (2100 clips) is a follow-up pending GPU-side evaluation.

Bibliography

[1] D. Mukhutdinov, A. Alex, L. Wang, and A. Cavallaro, "Deep Learning Models for Single-Channel Speech Enhancement on Drones," *IEEE Access*, vol. 11, pp. 22993–23007, 2023, doi: 10.1109/ACCESS.2023.3253719.